ME 5309 – Mid-Semester Examination: ANSWER SHEET

Roll No.: ME-MODEL **Name:** MODEL ANSWER SCRIPT

Questions attempted: Q1, Q2, Q3, Q4

Question 1

Given: sensor 1280 px wide, pixel pitch p = 5 µm (square), lens f = 16 mm, standoff Z = 400 mm; force sensor 10 mV/N, 12-bit ADC over 0–5 V; plant $\dot{x} = -2x + u$ with zero-order hold, T = 0.1 s.

(a) Assumptions: pinhole (thin-lens) camera model, lens focused on the belt plane, belt plane perpendicular to the optical axis, no lens distortion, so every pixel images the same length on the belt.

(i) The focal length in pixel units is the physical focal length divided by the pixel pitch: $f_x = f/p = (16\times10^{-3}\ \text{m})/(5\times10^{-6}\ \text{m}) = 3200$. The metres cancel so the answer is in pixels. **f_x = 3200 px**.

(ii) Width of the sensor = N·p = 1280 × 5 µm = 6400 µm = 6.4 mm. Using similar triangles between the sensor and the belt plane, $W_s/f = \text{FOV}/Z$, hence $\text{FOV} = W_s \cdot Z/f = (6.4\ \text{mm})(400\ \text{mm})/(16\ \text{mm}) = 160$ mm. **FOV = 160 mm**.

Ground sampling distance = FOV/N = 160 mm / 1280 px = 0.125 mm/px. Cross-check with the pixel-based formula Z/f_x = 400 mm / 3200 px = 0.125 mm/px; both agree, so **1 pixel = 0.125 mm = 125 µm** on the belt.

(b) Assumptions: ideal 12-bit ADC (4096 codes), the full 0–5 V range is used, no offset or gain error, and sensor noise is below one LSB so quantisation sets the limit.

Voltage step: 1 LSB = $V_{FS}/2^N = 5\ \text{V}/4096 = 1.2207\times10^{-3}$ V = 1.2207 mV.

The sensor maps force to voltage by V = S·F with S = 10 mV/N, so $\Delta F_{min} = \Delta V/S$ = 1.2207 mV / (10 mV/N) = 0.12207 N. **$\Delta F_{min} \approx 0.122$ N**. (Using 4095 steps instead of 4096 codes gives 0.1221 N, the same at 3 s.f.)

(c) With u held constant over [kT, (k+1)T] (zero-order hold) the variation-of-constants formula gives $x((k+1)T) = e^{AT}x(kT) + \int_0^T e^{A(T-s)}B\,u(kT)\,ds$. Since u(kT) is constant it leaves the integral, so $F = e^{AT}$ and $G = \int_0^T e^{A\sigma}B\,d\sigma = A^{-1}(e^{AT} - I)B$ for invertible A.

Substituting A = −2, B = 1, T = 0.1 s: $F = e^{-0.2} = 0.81873$. $G = (0.81873 - 1)/(-2) = 0.090635$.

Sanity check: the DC gain of the continuous plant is $-A^{-1}B = 0.5$, and the discrete DC gain $G/(1 - F) = 0.090635/0.18127 = 0.5$ ✓.

$x_{k+1} = 0.8187\,x_k + 0.0906\,u_k$

Question 2

Given: J = 0.02 kg·m², b = 0.05 N·m·s/rad, m = 1 kg, ℓ = 0.25 m, g = 9.81 m/s². So mgℓ = 1 × 9.81 × 0.25 = 2.4525 N·m (used throughout).

(a) The model is second order, so I choose the position and velocity as states: $x = [x_1, x_2]^T = [\theta, \dot{\theta}]^T$, with input τ.

Rewriting $J\ddot{\theta} + b\dot{\theta} + mg\ell\sin\theta = \tau$ gives the nonlinear state-space model: $\dot{x}_1 = x_2$, $\dot{x}_2 = (1/J)(\tau - b\,x_2 - mg\ell\sin x_1)$, i.e. $\dot{x} = f(x, \tau)$.

Equilibria for constant τ_0 need $\dot{x} = 0$: the first equation gives $x_2 = 0$, and the second then gives $mg\ell\sin\theta_e = \tau_0$, i.e. $\sin\theta_e = \tau_0/(mg\ell)$.

This has a solution only if $|\tau_0/(mg\ell)| \le 1$, so **$|\tau_0| \le mg\ell = 2.4525$ N·m**. For $|\tau_0| < mg\ell$ there are two equilibria per revolution (θ_e and $\pi - \theta_e$); at $|\tau_0| = mg\ell$ they merge, and beyond it the joint cannot be held statically.

(b) The equilibrium at $\theta_e = 60°$ needs the torque $\tau_e = mg\ell \sin 60° = 2.4525 \times 0.8660 = 2.124$ N·m. Define deviations $\delta x = x - x_e$ with $x_e = [60°, 0]^T$ and $\delta\tau = \tau - \tau_e$; then $\delta\dot{x} \approx A\,\delta x + B\,\delta\tau$ with the Jacobians evaluated at the equilibrium.

$\partial f_1/\partial x = [0, 1]$. $\partial f_2/\partial x_1 = -(mg\ell/J)\cos x_1 = -(2.4525/0.02)(0.5) = -122.625 \times 0.5 = -61.31$ (s^{-2}). $\partial f_2/\partial x_2 = -b/J = -0.05/0.02 = -2.5\ s^{-1}$.

A = [0, 1; −61.31, −2.5], and since $\partial f_2/\partial\tau = 1/J = 50$, **B = [0; 50]**.

Check: the characteristic equation $\lambda^2 + 2.5\lambda + 61.31 = 0$ gives $\lambda = -1.25 \pm 7.73j$, a lightly damped stable oscillation, as expected for a pendulum hanging near-vertical (60° below the downward vertical).

(c) Set all time derivatives to zero in the closed loop. With $\tau = K_p(\theta_d - \theta) - K_d\dot{\theta}$ and $\dot{\theta} = \ddot{\theta} = 0$ the damper term vanishes (so K_d has no effect on the final position) and the dynamics reduce to $mg\ell \sin\theta_{ss} = K_p(\theta_d - \theta_{ss})$.

Why $\theta_{ss} \neq \theta_d$: if the joint were at rest exactly at $\theta_d = 90°$ the PD torque would be $K_p \cdot 0 - K_d \cdot 0 = 0$, yet gravity pulls with $mg\ell \sin 90° = 2.4525$ N·m. The net torque would be non-zero, so the joint could not be at rest. A purely proportional-derivative law can only hold a load if there is a steady error.

Let $e = \theta_d - \theta_{ss}$ (in rad). Then $\sin\theta_{ss} = \sin(\pi/2 - e) = \cos e$, and the equation becomes $50\,e = 2.4525 \cos e$. Fixed-point iteration: $e_0 = 2.4525/50 = 0.04905$ (taking $\cos e \approx 1$); $e_1 = 0.04905 \times \cos(0.04905) = 0.04899$; $e_2 = 0.04905 \times \cos(0.04899) = 0.04899$, converged.

$e = 0.04899$ rad $\times (180/\pi) = 2.807°$, so $\theta_{ss} = 87.19°$. **Steady-state error ≈ 2.81° (the arm sags below the target).** This is small because K_p is much larger than $mg\ell$ ($K_p/mg\ell \approx 20$).

(d) Steady state: $\dot{\theta} = \ddot{\theta} = 0$ and the control law gives $\tau = K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss}$. Plugging into the plant, $mg\ell \sin\theta_{ss} = K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss}$, so $K_p(\theta_d - \theta_{ss}) = 0$ and $\boldsymbol{\theta_{ss} = \theta_d}$**: steady-state error is zero**.

Poles: linearise the closed loop about θ_d. Before compensation, $J\ddot{\theta} + (b + K_d)\dot{\theta} + (K_p + mg\ell\cos\theta_d)\,\delta\theta = 0$, characteristic polynomial $J\lambda^2 + (b + K_d)\lambda + K_p + mg\ell\cos\theta_d$. After compensation the added term $mg\ell\sin\theta$ cancels the plant's gravity term exactly, leaving $J\lambda^2 + (b + K_d)\lambda + K_p$.

The two differ only by $mg\ell\cos\theta_d$, and for $\theta_d = 90°$ this is $2.4525 \times 0 = 0$. So **the closed-loop poles are unchanged** at this operating angle. I should stress this is because $\cos 90° = 0$; if θ_d were, say, 60°, the compensation would raise the stiffness by 61.3×0.02... i.e. it would move the poles.

Question 3

Given: A = [0, 1; 0, −2], B = [0; 1], C = [1, 0], f_s = 1 kHz so the sample time is $T = 1/f_s$ = 1 ms.

(a) Reason 1 – noise amplification: the backward difference $\hat{\omega}_k = (\theta_k - \theta_{k-1})/T$ has frequency response $(1 - e^{-j\omega T})/T$, whose magnitude grows with frequency ($\approx \omega$ for small ωT), i.e. it is a high-pass operation. Any noise on the encoder signal therefore appears in the velocity multiplied by roughly $1/T = 1000\ s^{-1}$, and can swamp the true velocity.

Reason 2 – quantisation: the encoder only reports integer counts, so $\Delta\theta$ is an integer multiple of the count size q. Dividing by T = 1 ms gives velocity steps of q/T; for a 4096-count encoder $q = 2\pi/4096 = 1.53\times10^{-3}$ rad, so one step is 1.53 rad/s. At low speed the estimate flips between 0 and one step (a noisy staircase), exactly where the damping term of the PD law is needed.

(A further, smaller issue is the half-sample delay of a backward difference, but the two reasons above are the main ones.)

(b) The analogue prototype is the first-order low-pass $H(s) = \omega_c/(s + \omega_c)$ with $\omega_c = 2\pi f_c = 2\pi \times 20 = 125.66$ rad/s. The bilinear (Tustin) transform replaces s by $(2/T)(z - 1)/(z + 1)$ with $T = 1/f_s = 0.001$ s.

Define $a = \omega_c T/2 = 125.66 \times 0.001/2 = 0.062832$. Then $H(z) = \omega_c / [(2/T)(z-1)/(z+1) + \omega_c] = a(z+1) / [(1+a)z - (1-a)] = a(1+z^{-1}) / [(1+a) - (1-a)z^{-1}]$.

Dividing through by $(1+a)$: $a/(1+a) = 0.062832/1.062832 = 0.059117$ and $(1-a)/(1+a) = 0.937168/1.062832 = 0.881767$. Hence with u = differenced signal and y = filtered velocity: $\mathbf{y_k = 0.05912\,(u_k + u_{k-1}) + 0.88177\,y_{k-1}}$. (Frequency pre-warping is neglected since $f_c \ll f_s/2$.)

Verification of the DC gain: set u and y constant ($z = 1$): $y = 2(0.05912)u + 0.88177\,y \Rightarrow y(1 - 0.88177) = 0.11824\,u \Rightarrow y/u = 0.11824/0.11823 = 1.000$. In general $H(1) = 2a/[(1+a) - (1-a)] = 2a/2a = 1$ exactly, so the steady-state output equals the commanded input ✓.

(c) First check observability: $O = [C; CA] = [1, 0; 0, 1]$, full rank, so the observer poles can be placed arbitrarily.

With $L = [\ell_1, \ell_2]^T$ and $C = [1, 0]$, $LC = [\ell_1, 0; \ell_2, 0]$, so $A - LC = [-\ell_1, 1; -\ell_2, -2]$.

Characteristic polynomial: $\det(sI - (A - LC)) = \det[s + \ell_1, -1; \ell_2, s + 2] = (s + \ell_1)(s + 2) - (-1)(\ell_2) = s^2 + (2 + \ell_1)s + (2\ell_1 + \ell_2)$.

Both estimator poles at $s = -10$ means the desired polynomial is $(s + 10)^2 = s^2 + 20s + 100$. Comparing coefficients: s^1: $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$. s^0: $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 100 - 36 = 64$.

$\mathbf{L = [18, 64]^T}$. Verification: $A - LC = [-18, 1; -64, -2]$ has trace -20 and determinant $(-18)(-2) - (1)(-64) = 36 + 64 = 100$, so the polynomial is $s^2 + 20s + 100 = (s + 10)^2$ ✓.

(d) Advantage of the observer that no filter of this kind can give: the observer is driven by both the measurement y and the model with the actual input u, so it predicts the motion between samples. The velocity estimate therefore has little or no phase lag. The low-pass filter on differenced data, by contrast, always pays a lag of about $1/(2\pi f_c) = 8$ ms for its noise rejection, a trade-off it cannot escape. In addition the observer gives all states, and can be augmented with a disturbance state (e.g. unknown load torque) which no filter on θ can estimate.

Circumstance where the filter is safer: when the plant model is poorly known or changes during operation, for example an unknown payload, varying friction, contact with the environment, or actuator saturation making u differ from the commanded value. The observer propagates the wrong model so its estimate can be biased (and in the worst case unstable with the feedback loop), whereas the model-free filter just smooths what the encoder actually measures.

Question 4

Given: plant $\ddot{\theta} + a\dot{\theta} + c\theta = b\,u$ with $a = 2$, $c = 6$, $b = 1$; controller $u = N_{ff}\,r - k_1\theta - k_2\dot{\theta}$.

(a) Substitute the control law into the plant: $\ddot{\theta} + a\dot{\theta} + c\theta = b\,N_{ff}\,r - b\,k_1\,\theta - b\,k_2\,\dot{\theta}$. Moving the feedback terms to the left: $\ddot{\theta} + (a + b\,k_2)\,\dot{\theta} + (c + b\,k_1)\,\theta = b\,N_{ff}\,r$.

Numerically, $a + b\,k_2 = 2 + 2 = 4$ and $c + b\,k_1 = 6 + 2 = 8$, so the characteristic equation is $s^2 + 4s + 8 = 0$. Roots: $s = [-4 \pm \sqrt{(16 - 32)}]/2 = -2 \pm 2j$. They have negative real parts, so the loop is stable; $\omega_n = \sqrt{8} = 2.83$ rad/s and $\zeta = 4/(2 \times 2.83) = 0.707$.

Transfer function from r to θ: $T(s) = b\,N_{ff} / (s^2 + 4s + 8)$. DC gain ($s = 0$): $T(0) = N_{ff}/8$. For unity DC gain, $\mathbf{N_{ff} = 8}$. (Note that this is larger than 1 because the gravity-related stiffness c and the feedback k_1 both resist the displacement.)

(b) With the actual $b = 0.75$ and unchanged $N_{ff} = 8$, $k_1 = k_2 = 2$, the closed loop becomes $\ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\,r$, i.e. $\ddot{\theta} + 3.5\dot{\theta} + 7.5\theta = 6\,r$.

For a unit step the steady state is found by setting derivatives to zero: $7.5\,\theta_{ss} = 6 \times 1$, so $\theta_{ss} = 6/7.5 = \mathbf{0.8}$.

Tracking error $= (r - \theta_{ss})/r \times 100\% = (1 - 0.8) \times 100 = \mathbf{20\%}$. A common mistake would be to say the output falls by the same 25% as the actuator gain; that is wrong here, because b also appears in the feedback stiffness $c + bk_1$, which decreases from 8 to 7.5 and partially compensates. The poles also change: $s^2 + 3.5s + 7.5 \rightarrow s = -1.75 \pm 2.11j$, still stable.

(c) Take the state $z = [\theta, \dot{\theta}, \xi]^T$. The plant gives $\ddot{\theta} = -a\dot{\theta} - c\theta + b\,u$ with $u = -k_1\theta - k_2\dot{\theta} + K_I\xi$, and $\dot{\xi} = r - \theta$. With $r = 0$ (r only shifts the equilibrium) the closed-loop matrix is $A_{cl} = [0, 1, 0; -(c + bk_1), -(a + bk_2), bK_I; -1, 0, 0]$.

Characteristic polynomial $\det(sI - A_{cl})$, expanding along the first row: $s\cdot[(s + a + bk_2)s - 0] + 1\cdot[(c + bk_1)s + bK_I] = s^3 + (a + bk_2)s^2 + (c + bk_1)s + bK_I$. **(Result)**

Desired poles -4 and $-2 \pm 2j$: $(s + 2 - 2j)(s + 2 + 2j) = (s + 2)^2 + 4 = s^2 + 4s + 8$. Multiply by $(s + 4)$: $s^3 + 4s^2 + 8s + 4s^2 + 16s + 32 = s^3 + 8s^2 + 24s + 32$.

Compare coefficients with $a = 2$, $b = 1$, $c = 6$: s^2: $2 + k_2 = 8 \Rightarrow \mathbf{k_2 = 6}$. s^1: $6 + k_1 = 24 \Rightarrow \mathbf{k_1 = 18}$. s^0: $b\,K_I = 32 \Rightarrow \mathbf{K_I = 32}$.

Check: all coefficients of the desired polynomial are positive and $(8)(24) = 192 > 32$, so Routh–Hurwitz confirms the three poles are in the left half plane ✓.

(d) Suppose the closed loop is stable and r is constant. Then the state converges to a constant, so $\dot{\theta} = \ddot{\theta} = 0$ and, crucially, $\dot{\xi} = 0$ as well (ξ itself is constant at its final value ξ_{ss}). From the integrator equation $\dot{\xi} = r - \theta$, we get $r - \theta_{ss} = 0$, i.e. $\boldsymbol{\theta_{ss} = r}$.

Neither b nor c appears in this argument. They only influence the value of ξ_{ss} needed to hold the equilibrium (from the plant equation $c\theta_{ss} = bu_{ss} = b(-k_1\theta_{ss} + K_I\xi_{ss})$), and the pole locations, so the zero error survives any error in the model parameters provided the loop stays stable (unlike the feed-forward N_{ff} scheme of part (b)).

Practical failure mode: integrator windup. If the actuator saturates (torque limit), the loop is effectively open, the error stays non-zero and ξ keeps integrating to a huge value. When the reference becomes reachable the stored integral forces a large overshoot and a long recovery. Standard remedy: anti-windup, such as clamping the integrator when the actuator is saturated, conditional integration, or back-calculation that feeds the difference between commanded and saturated torque back to the integrator.